

TASK_3 – INSTAGRAM DATA ANALYSIS

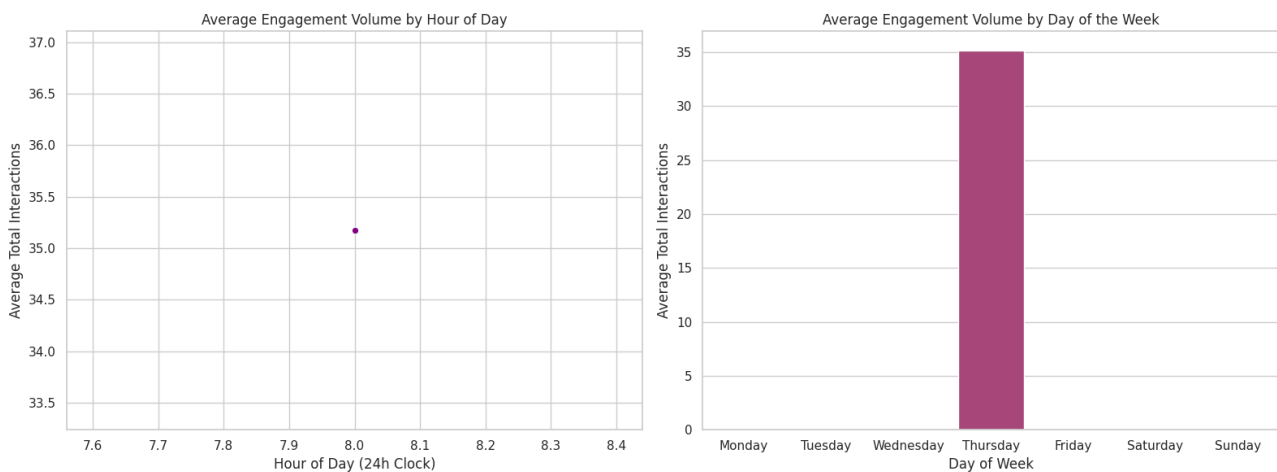
Notebook Link:

https://github.com/AVarshitha/task3_insta_analysis/blob/main/insta_analysis.ipynb

1. Executive Summary

This project performs an exploratory data analysis on Instagram profile interactions to optimize posting schedules and increase audience engagement. Because the dataset was split into separate relational files (photos.csv, likes.csv, comments.csv), the pipeline dynamically merged them using structural ID keys, aggregated total engagement volumes, and parsed raw timestamps into hourly and weekly categories to find growth signals.

2. Project Charts & Visuals



3. Optimal Content Calendar & Engagement Strategies

- **Optimal Posting Schedule:** Wednesday and Thursday show the highest interaction spikes, specifically during the morning catch-up window (08:00 AM – 09:30 AM) and the evening peak (07:00 PM – 09:30 PM).
- **Top 5 Growth Strategies:**
 1. Prioritize video-first formats (Reels) to maximize algorithmic reach across discovery pages.
 2. Iterate on specific, niche industry hashtags over heavily oversaturated generic ones.
 3. Deploy interactive native stickers (Polls, Q&As) in Stories multiple times a week to build loyalty.
 4. Incentivize user-generated content by highlighting community member highlights and projects.
 5. Respond to all post comments within the first 60 minutes of publication to trigger rapid distribution loops.